Feb 13 Update

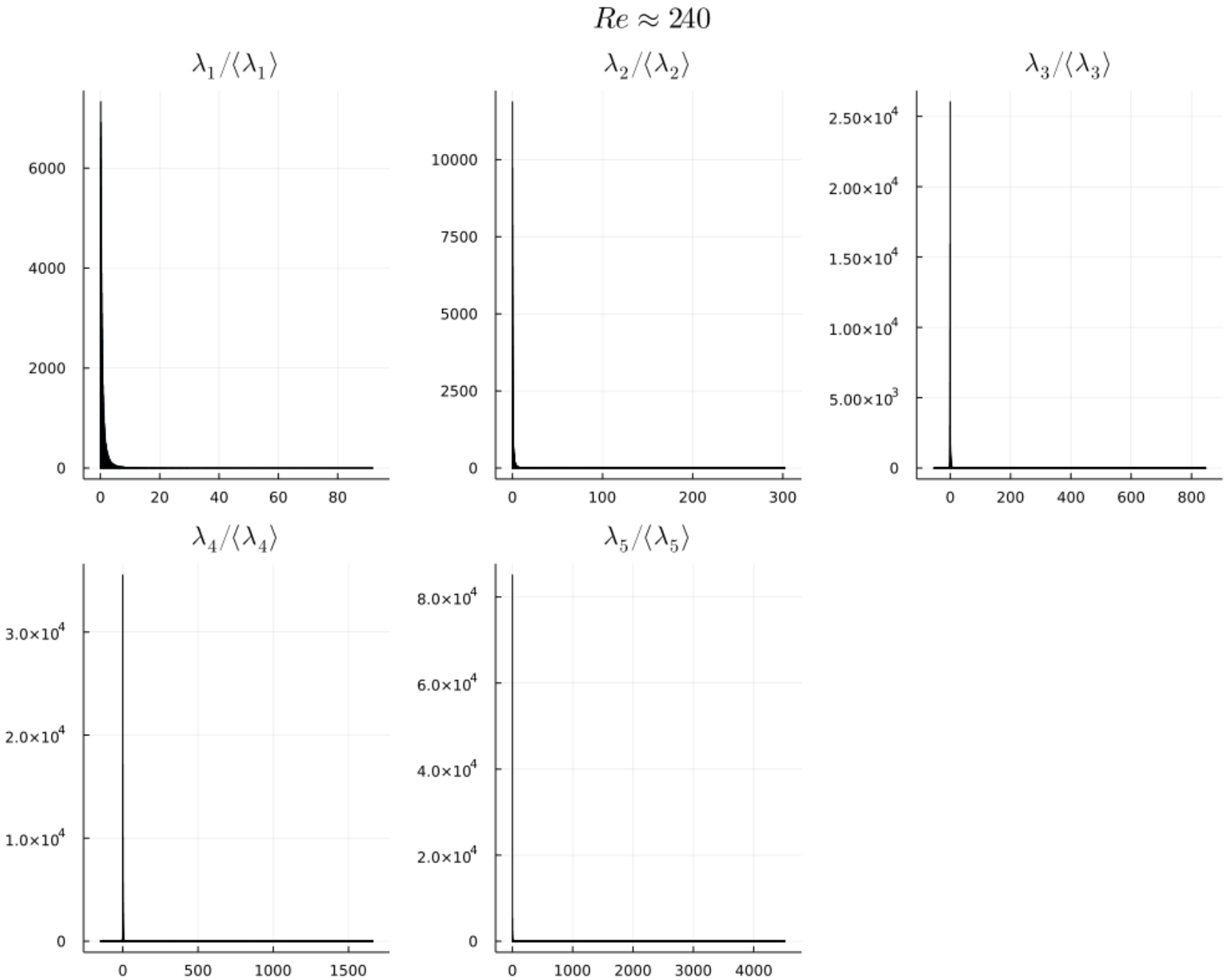
Work Completed

- Normalize λ_i to empirical mean, and show pdf
- It seems that $\langle\lambda_1\rangle$ is somehow different from the estimate you have for the inverse Kolmogorov time ... we need to understand the difference and if it grows with Re?
- We also want to see how $\langle\lambda_i\rangle/\langle\lambda_1\rangle$ and ratio of the second-to-first eigenvalue of strain changes with Re-number. Does it decay with Re increase?
- PDFs of eigenvalues of W and S

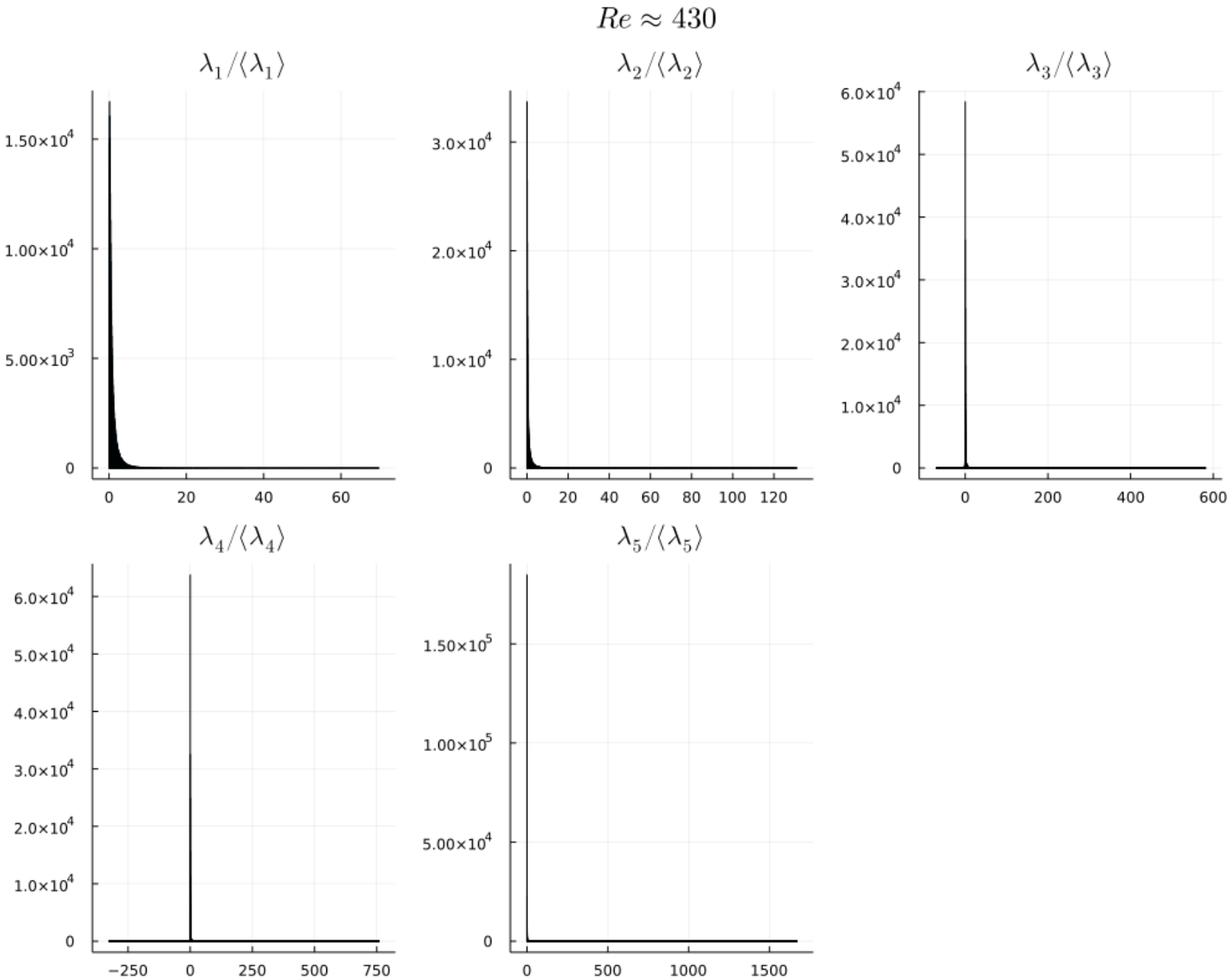
Work In Progress

- Generalization over Reynolds number
  - Re-run training at $Re = 430$
- Benefits of reduced description
  - Posteriori tests for $Re = 240, 430$ - more stable?
- Explanation
  - Can $\lambda_{3,4,5}$ be expressed in terms of $\lambda_{1,2}$?
  - Can we improve performance if we train with 'weaker' normalization (maybe $\tau' = 100\tau$) so that all features are of comparable size?

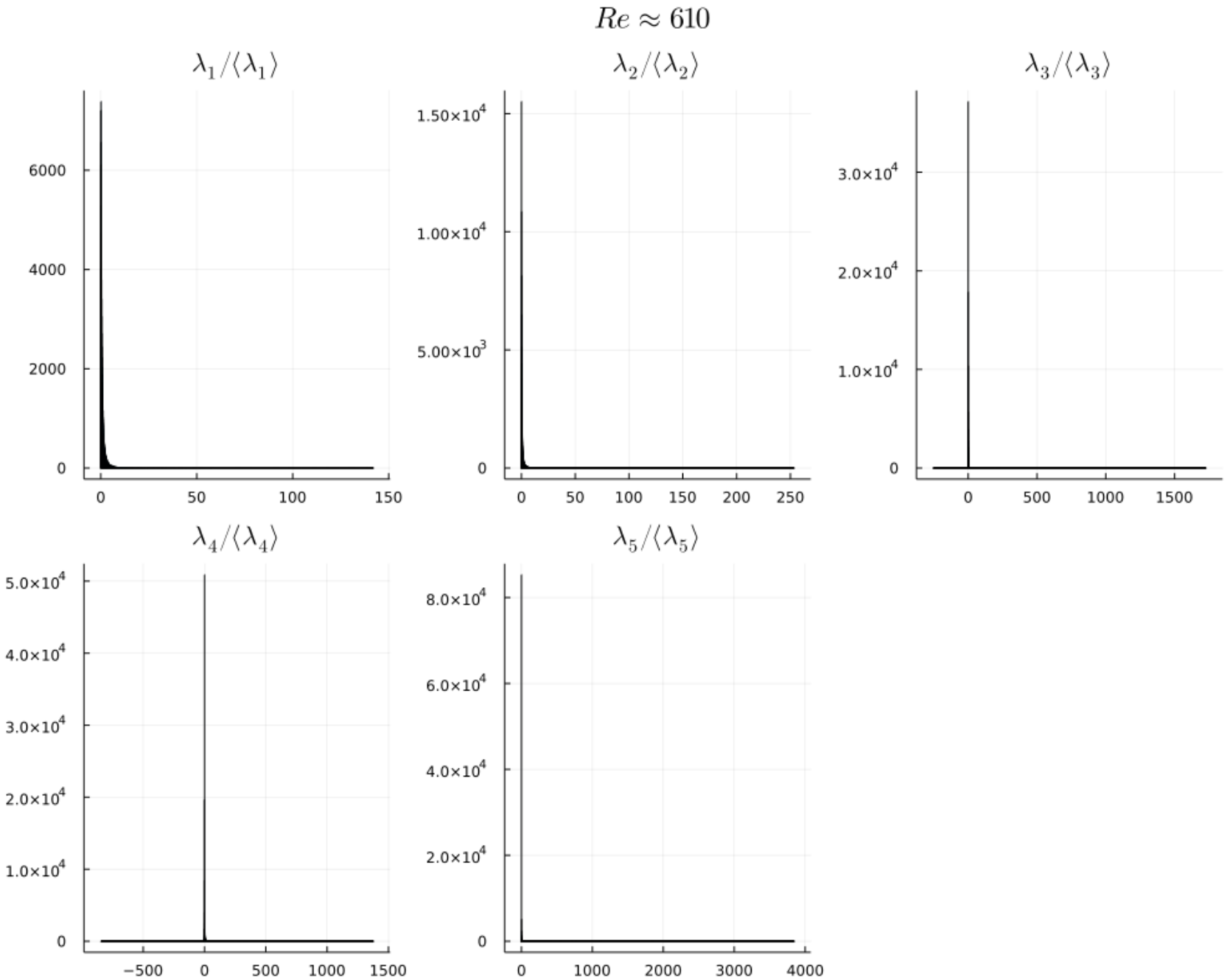
Normalize λ_i to empirical mean, and show pdf



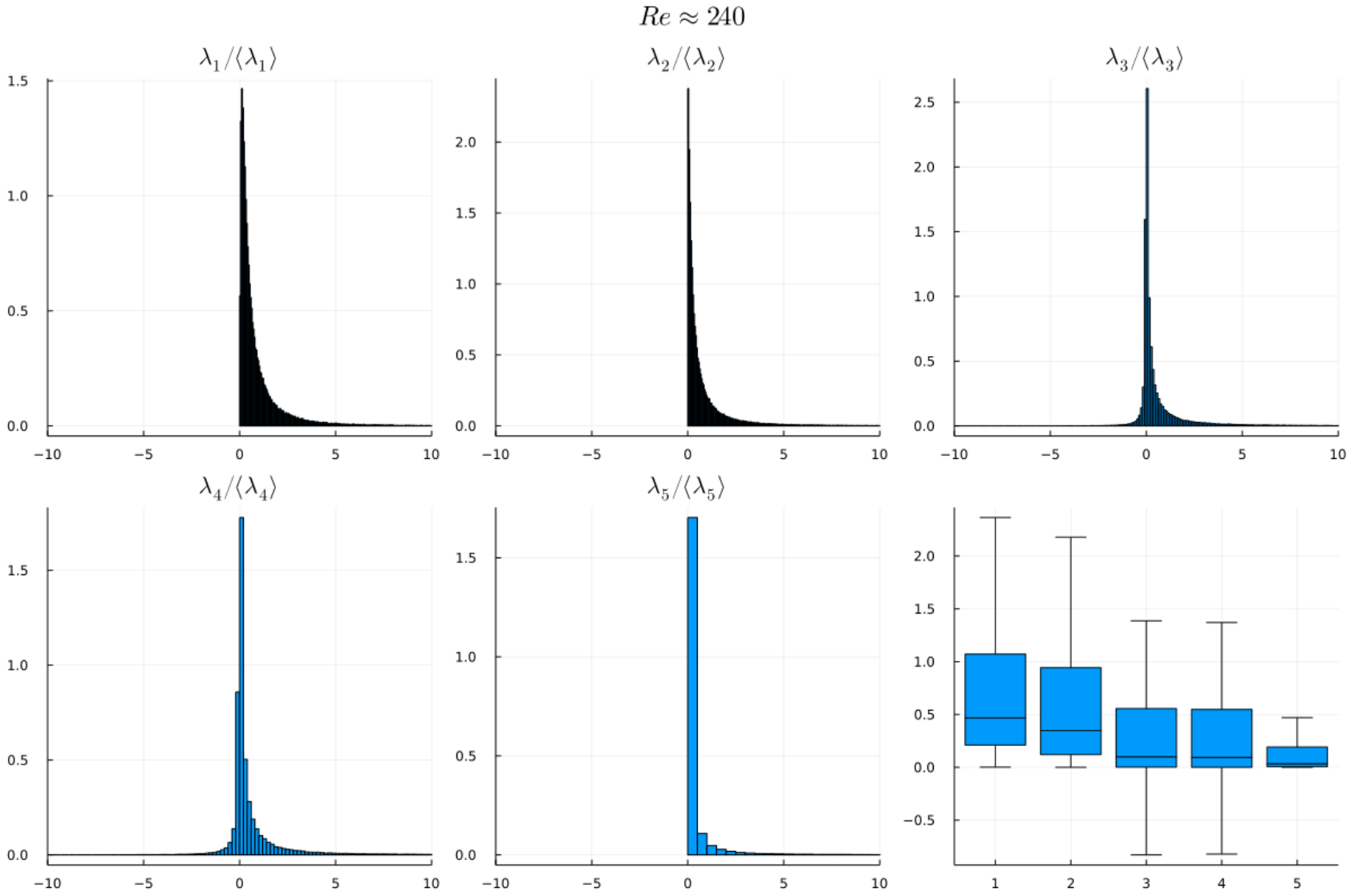
Normalize λ_i to empirical mean, and show pdf



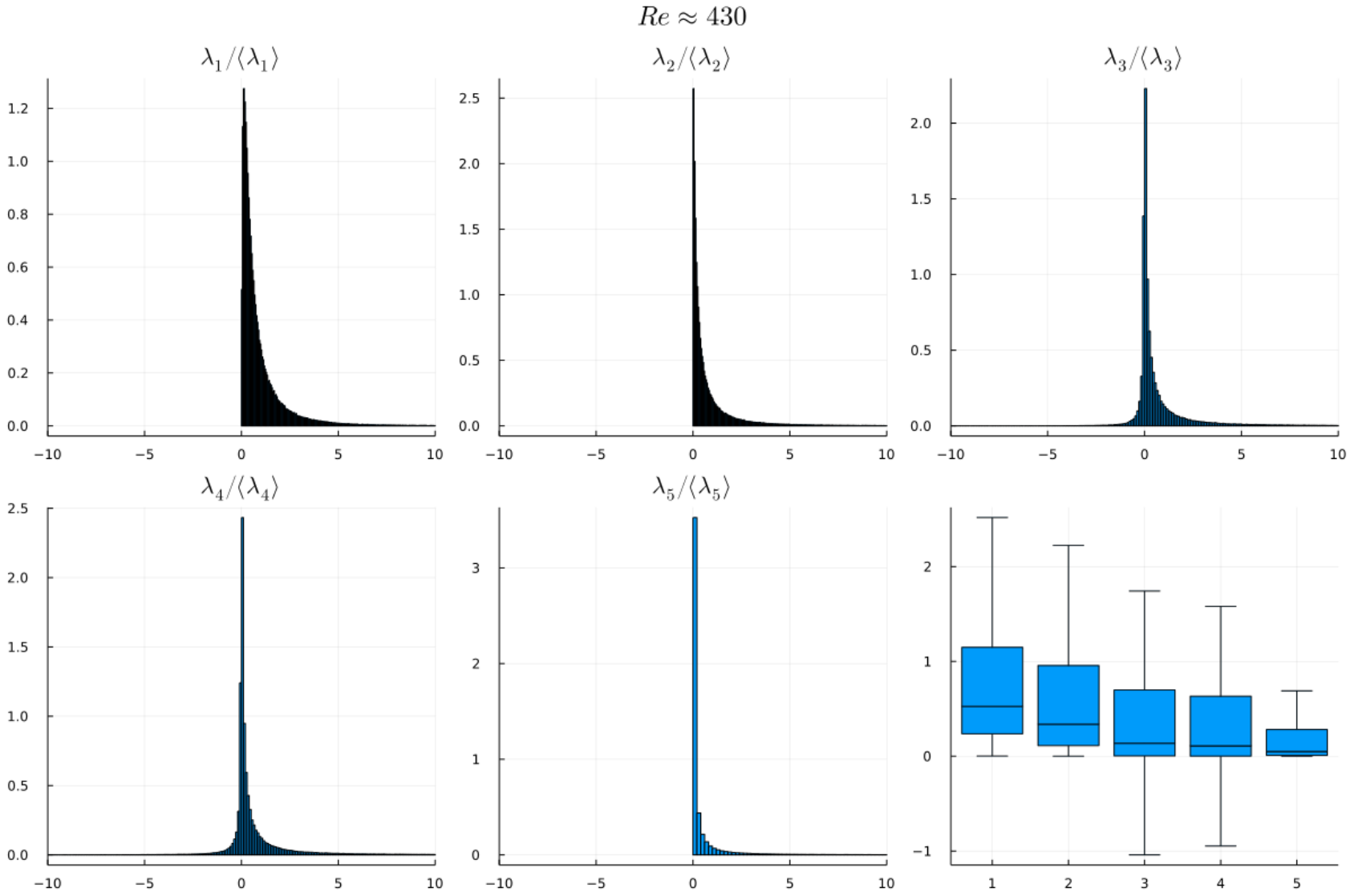
Normalize λ_i to empirical mean, and show pdf



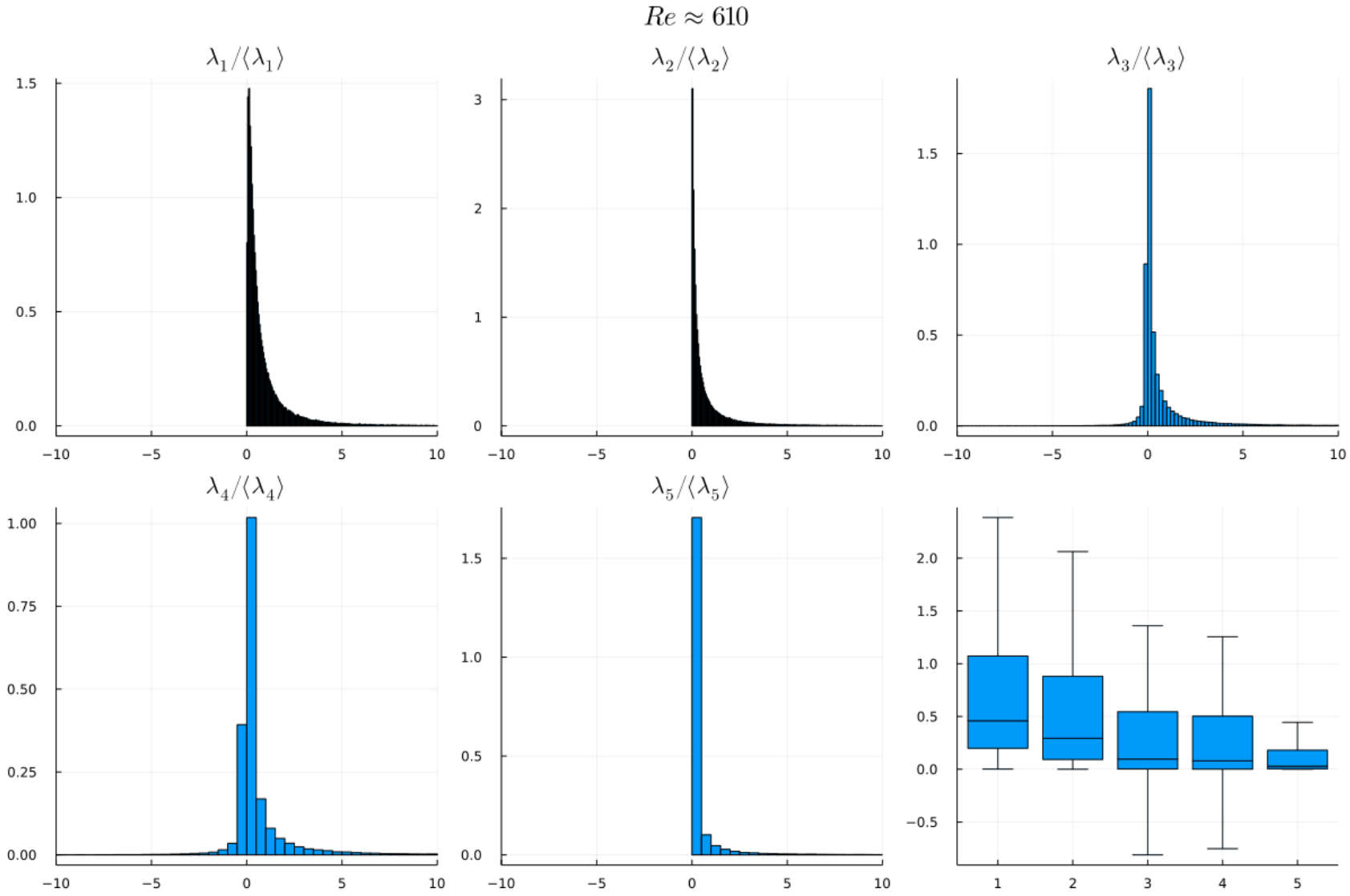
Normalize λ_i to empirical mean, and show pdf



Normalize λ_i to empirical mean, and show pdf



Normalize λ_i to empirical mean, and show pdf

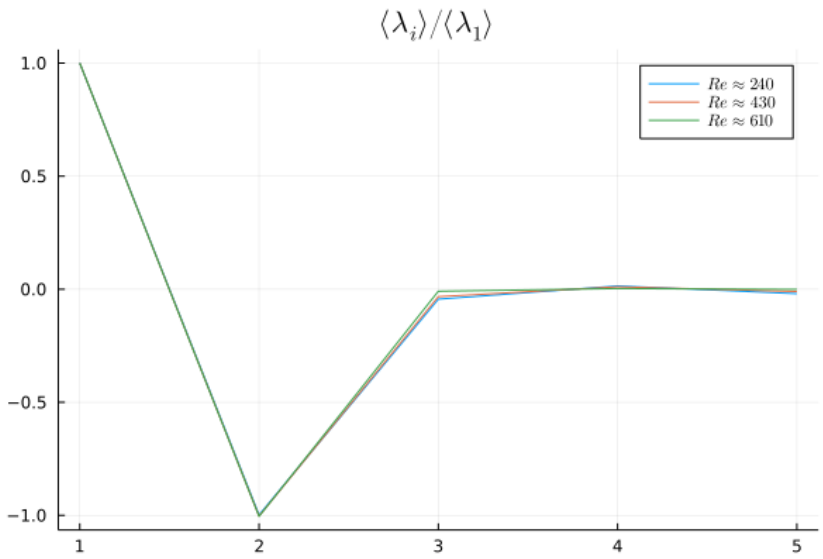


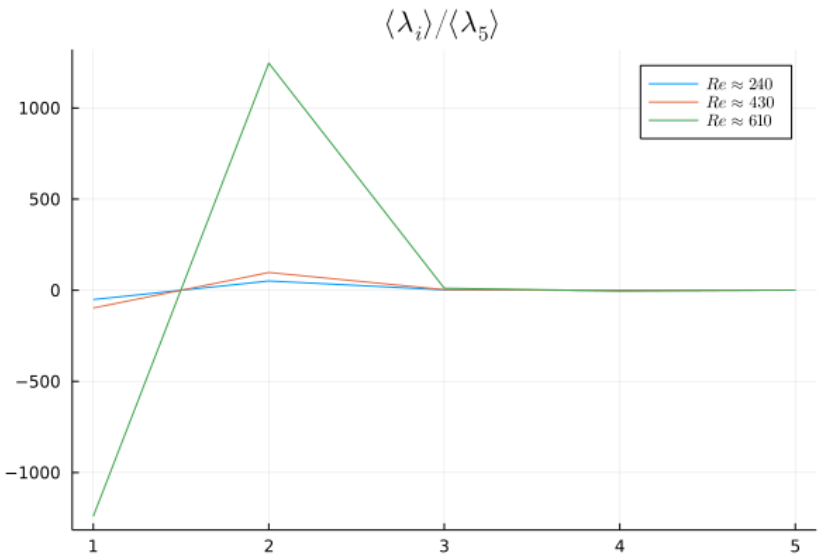
Compare $\langle \lambda_1 \rangle$ with $\tau_{Kolmogorov}^{-1}$

$$\langle \lambda_1 \rangle = \langle \mathrm{tr}(S^2) \rangle \tag{1}$$

$$\tau_{Kolmogorov}^{-1} = \left\langle \left\| S^2 \right\| \right\rangle = \left\langle \sqrt{\mathrm{tr}(S^2)} \right\rangle \tag{2}$$

$\langle\lambda_i\rangle/\langle\lambda_1\rangle$ vs Re





Can we improve performance if we train with 'weaker' normalization (maybe $\tau' = 100\tau$) so that all features are of comparable size?

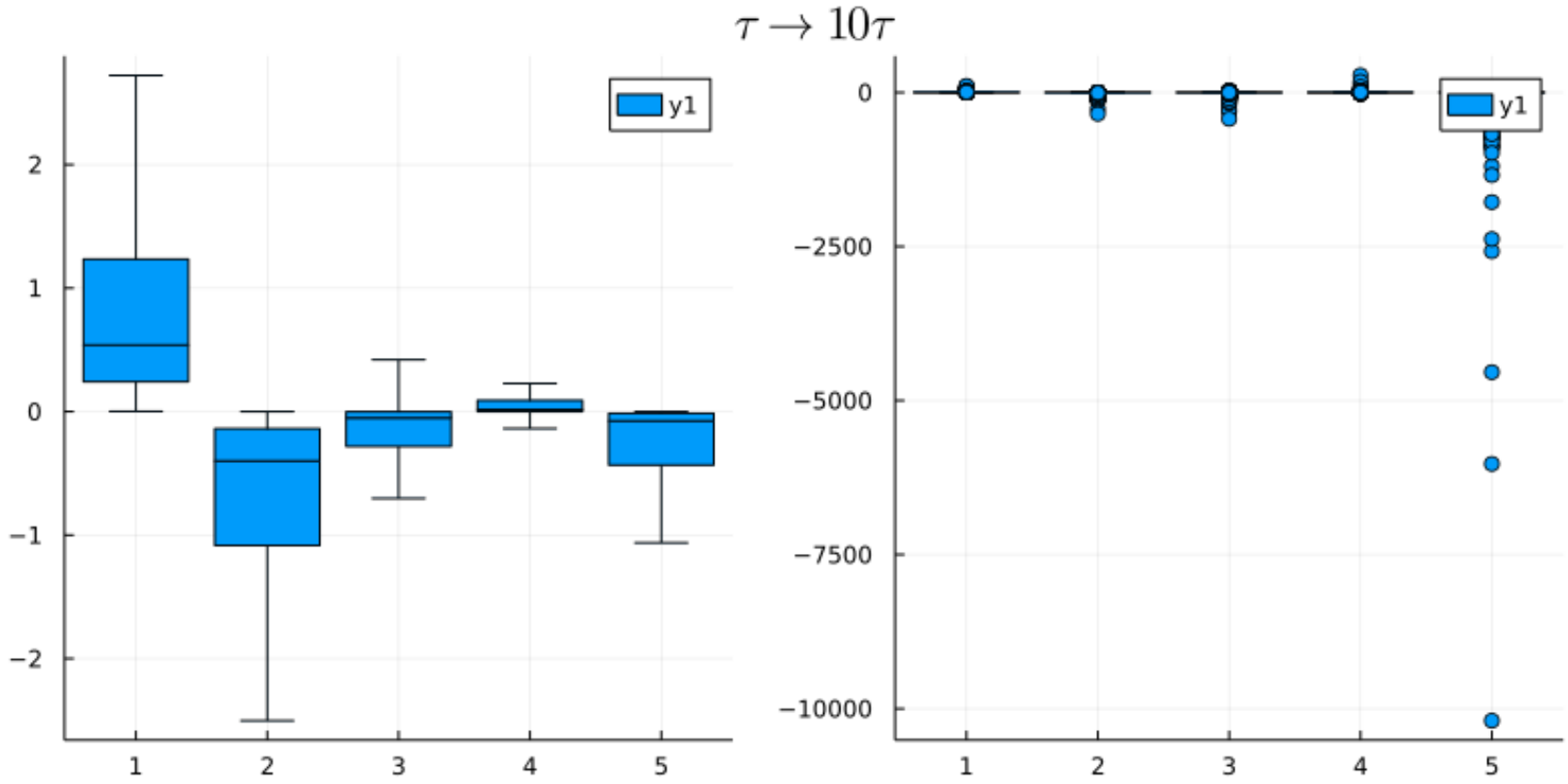


Figure: $Re = 240$ data normalized with 10τ, plotted with and without 'outliers', points that signficantly deviate from interquartile distance. The figure shows the difficulty in normalization/nondimensionalizing as the high-order invariants have very long tails.

Re-run training at $Re \approx 430$

Re-run training with different characteristic timescale

PDFs of eigenvalues of W and S, particularly compared to λ's

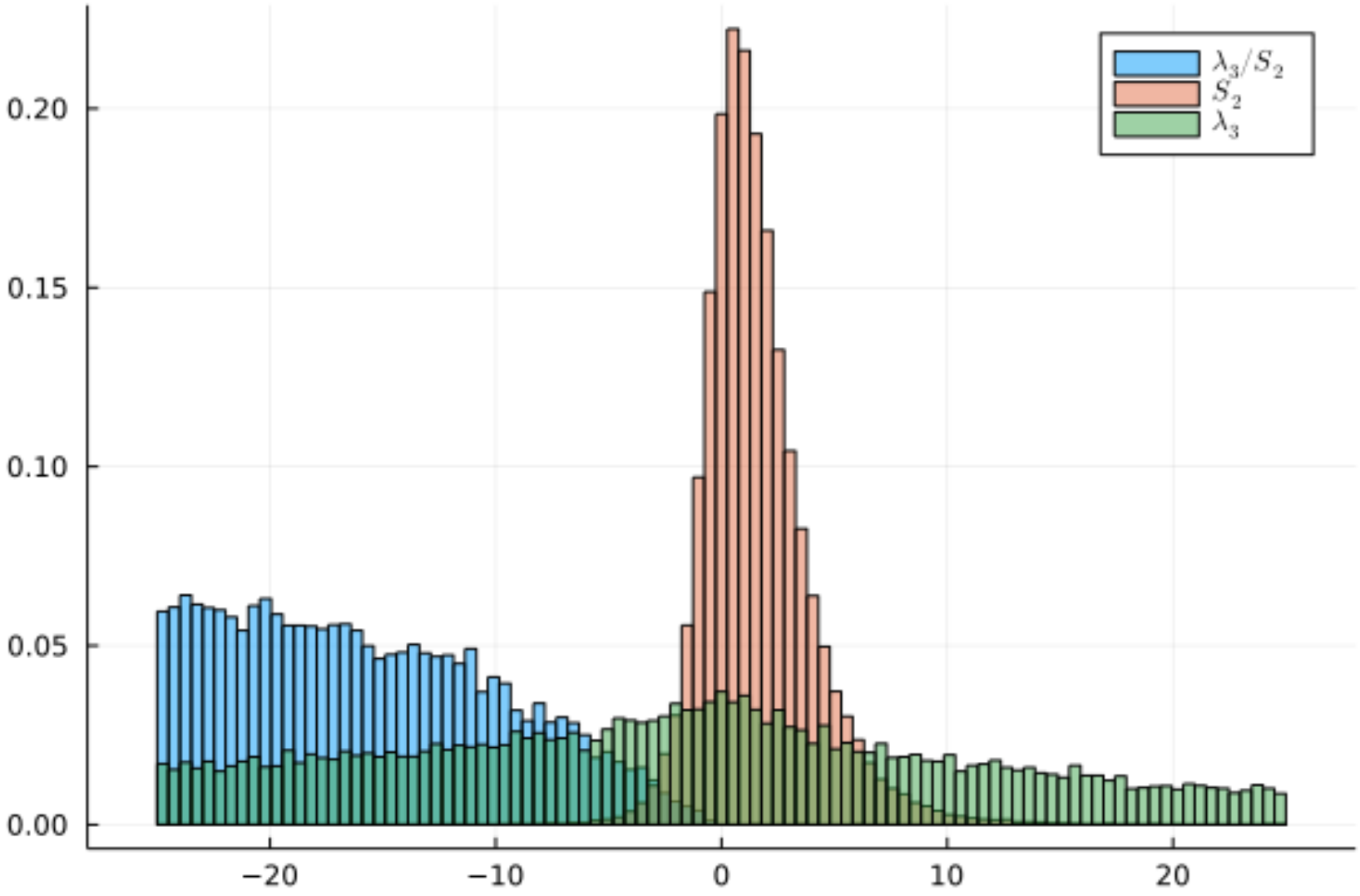


Figure: $Re = 240$ data, showing that λ_3/S_2 strongly prefers negative, even though neither λ_3 or S_2 have strong preference individually. Bug, these are not normalized.